Supplementary Material: Fixed Points and Certified Boundedness of Unrolled Born/Contrast-Source Inverse Scattering Networks

Longgang Wang, Guoqiang Wang, and Nianfeng Shi

1 Contrast-solve truncation sweep

The contrast step uses a fixed $n_{\mathrm{cg}} = 15$-step conjugate-gradient loop. The certified blocks C and D are closed-form and touch no iterative solve; the contrast-path blocks A, B and the full-layer norm that contains them do.

With the implemented n_{cg} the relative residual of the regularised system is 2.90×10^{-4}; it falls to 8.18×10^{-10} at $n_{\mathrm{cg}} = 200$. The field block D contains no iterative solve, and there the analytic directional derivative and a central difference agree to 8.6×10^{-13}. On the contrast-path block A at the implemented truncation the central difference and the analytic directional derivative disagree by 0.82; the relative discrepancy does not shrink as the finite-difference step is refined, which is the signature of a difference quotient that does not converge to the derivative at that truncation. Converging the solve restores the agreement to 1.79×10^{-2}.

Table 1: Contrast-solve truncation sweep at a fixed operating point, $n_{\mathrm{cg}} \in \{5, \ldots, 200\}$, float64. C, D are bit-identical over the whole sweep; the contrast-path blocks A, B and the full-layer norm that contains them are not, and their min and max are tabulated.

quantity	min	max	behaviour over the sweep
$\sigma_{\max}(C)$ (CG-free)	0.8480	0.8480	bit-identical
$\sigma_{\max}(D)$ (CG-free)	0.7935	0.7935	bit-identical
$\sigma_{\max}(A)$ (contrast)	0.723	0.874	varies
$\sigma_{\max}(B)$ (contrast)	0.118	3.515	varies
$\sigma_{\max}(DT_k)$ (full layer)	0.7368	3.6769	varies, non-monotonically

The full-layer norm mixes the two CG-free blocks with the two solver-dependent ones, so it inherits the artefact even though its own closure has no solver in it. It is reported because a reader who wants one number per layer will reach for it, and it is never certified: a statement about $\sigma_{\max}(DT_k)$ would be a statement about the truncation the trained weights happen to be calibrated to.

2 First-layer gradient norm

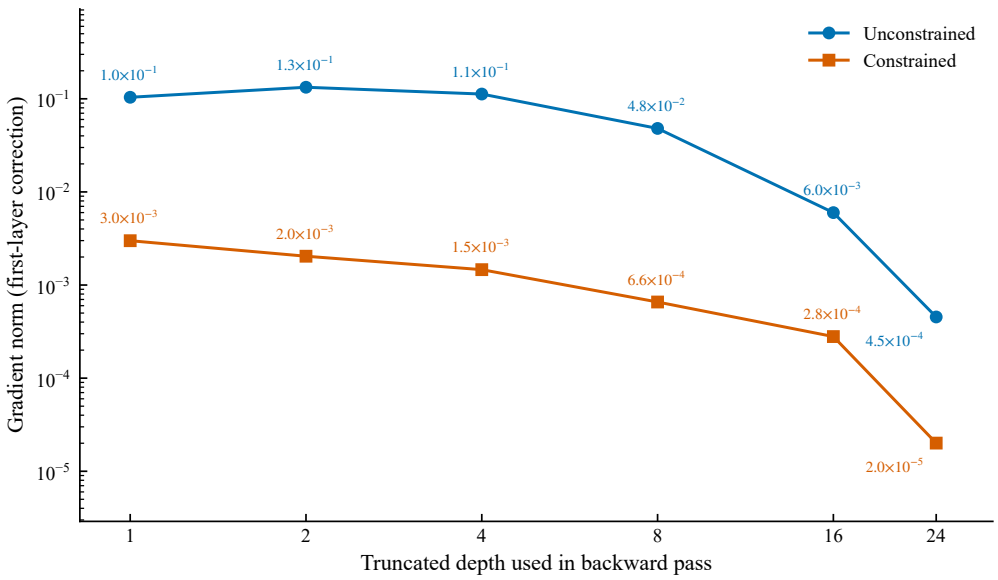


Figure 1: First-layer gradient norm versus the truncated depth used in the backward pass.

3 Driver provenance for Table ??

The block notation is that of Lemma 4.1 in the main text; the *Proved*/*Measured* tags carry the meaning stated in the table caption. Contrast-path rows (A, B, layer norm) are quoted at a *converged* contrast solve, with the as-implemented n_{cg} values listed separately in the main text (convergence control); the field quantities $\mathcal{E}$ and α_c are read at that same converged solve, i.e. from the n_{cg}-sweep driver, which is a different driver and contrast sample from the grid study. The composite rows (ρ_χ, ρ, M_J, the gap, composite $\sigma_{\max}$) are evaluated at the *trained per-layer* step sizes from the double-precision composite audit, a different operator from the per-layer rows. Row 1 is one quantity shown in two columns, because the design value and the certificate are the same expression, and it is not two independent checks. The Λ row quotes only the *field* term; with $\prod_i \|W_i\|_2 \le 1$ from the spectral clamp the omitted additive term is at most $\|\mathbf{G}_s\|_2 \bar{\epsilon}/s_{\mathrm{res}}$, i.e. a small fraction of the field term, so the tabulated value is the field term and the true bound is the sum.